Project: Worked examples
Title: Duct sizing
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates duct sizes by the equal-friction or the velocity method, rounded up to standard sizes. It reports the velocity, friction rate and losses of each section and the pressure drop along the index run.

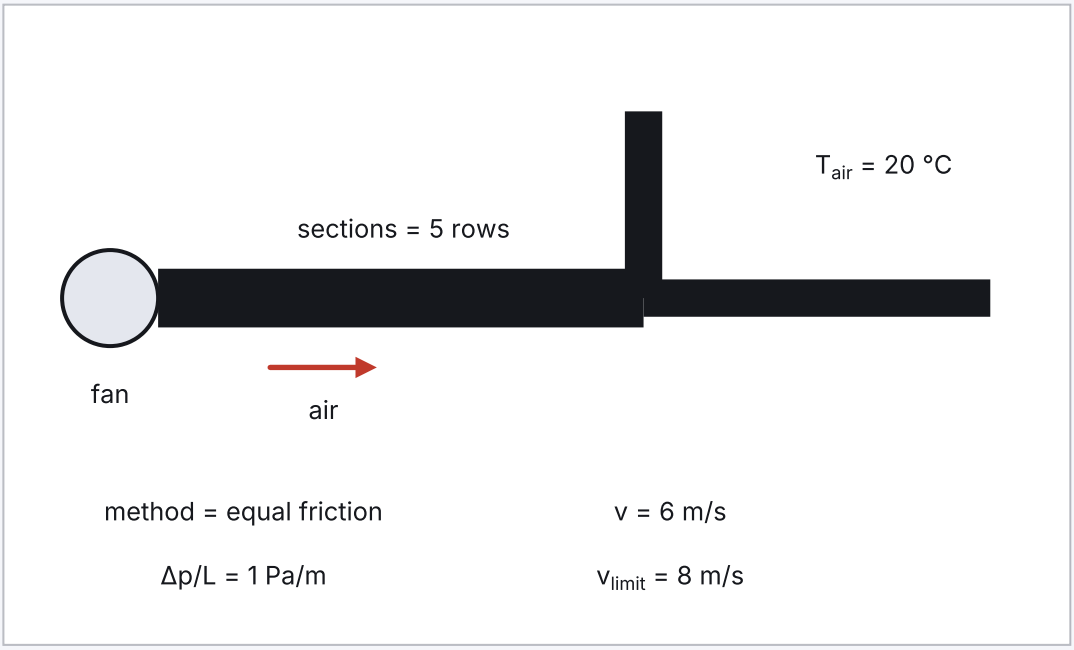


Figure 1 — A duct run from the fan through its sections as listed (a main and two branches drawn), each sized by the chosen method. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Round duct diameter from the design target	D by Colebrook with the air properties at T (equal friction), or D = √(4Q/πv) (velocity), rounded up to the next standard size
Rectangular duct of equal friction and capacity	D_e = 1.30 (ab)^0.625/(a + b)^0.25
Flat-oval duct of equal friction and capacity	D_e = 1.55 A^0.625/P^0.25, with A = πb²/4 + b(a − b) and P = πb + 2(a − b)
Velocity method, rectangular or flat-oval duct with side a fixed	b from the free area A = Q/v (rectangle b = A/a; flat oval the smaller root of (1 − π/4)b² − ab + A = 0), rounded up; D_e is kept for the friction only
Friction loss per metre, at the velocity v_e of the equivalent round duct	Δp/L = f ρ v_e^2/(2D_e)
Fitting losses, at the actual velocity	Δp = ΣC ρ v²/2
Index-run totals, summed over every section of the run	Δp_{run} = Σ(Δp/L · L + ΣC ρ v²/2)

Basis: Duct design (friction chart basis, equivalent diameters, loss coefficients)[1](Ch. 21); Flow of fluids in pipes and ducts[2](§4); Ventilation for buildings: sheet metal air ducts and fittings with rectangular cross-section, dimensions[3]; Ventilation for buildings: sheet metal air ducts and fittings with circular cross-section, dimensions[4].

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
method	Sizing method	equal friction		assumption: Equal friction for a low-pressure system.	any
sections	Duct sections	5 row(s)		drawing: DRG-M-205 rev B	any
$round_{to}$	Round sizes up to a multiple of	50	mm	spec: BS EN 1505 / 1506 preferred sizes, 50 mm steps	1 … 100 mm
eps_{rough}	Duct roughness	0.09	mm	spec: Galvanised sheet, medium roughness (ASHRAE Ch. 21 category)	0.03 … 3 mm
T_{air}	Air temperature	20	°C	assumption: Supply air 20 °C.	-20 … 60 °C
p_{atm}	Barometric pressure	101.325	kPa	assumption: Sea level.	80 … 105 kPa
dp_{target}	Target friction rate (equal friction)	1	Pa/m	spec: SPEC-M-07 §5: 1 Pa/m design friction rate	0.3 … 5 Pa/m
v_{target}	Target velocity (velocity method)	6	m/s	assumption: Velocity method only; equal friction does not use it, so the template default is kept.	2 … 20 m/s
v_{limit}	Velocity limit (warning and check)	8	m/s	spec: SPEC-M-07 §5: 8 m/s in risers	2 … 30 m/s
dp_{limit}	Total pressure-drop limit (check)	250	Pa	spec: Fan external static allowance 250 Pa	50 … 2000 Pa

TABLE ROW	VALUES
sections[1]	name = AHU to riser; Q = 2 m³/s; L = 15 m; shape = rectangular; a = 800 mm; D = —; C = 0.8; dp_eq = 0 Pa; index = yes
sections[2]	name = riser; Q = 2 m³/s; L = 12 m; shape = round; a = —; D = —; C = 0.3; dp_eq = 0 Pa; index = yes
sections[3]	name = floor main; Q = 1.2 m³/s; L = 25 m; shape = rectangular; a = 500 mm; D = —; C = 1.1; dp_eq = 0 Pa; index = yes
sections[4]	name = branch to VAV; Q = 0.4 m³/s; L = 10 m; shape = round; a = —; D = —; C = 0.9; dp_eq = 60 Pa; index = yes

TABLE ROW	VALUES
sections[5]	name = other branch; Q = 0.8 m³/s; L = 18 m; shape = round; a = —; D = —; C = 1; dp_eq = 60 Pa; index = no

3. Assumptions

A1. Air is dry at the stated temperature and pressure. (Ideal gas, ρ = p/(287.055 T).)
A2. The viscosity of air follows Sutherland's law. (Standard air properties.)
A3. Rectangular and flat-oval ducts are sized through the equivalent diameter for equal friction and capacity. (Huebscher / Heyt–Diaz relations as given in ASHRAE Fundamentals Ch. 21.)
A4. Fitting loss coefficients C are user-supplied (manufacturer or ASHRAE/CIBSE data, not embedded) and refer to the section's velocity pressure. (Non-negotiable: no standard tables embedded.)
A5. method (Sizing method) is taken as equal friction. (Equal friction for a low-pressure system.)
A6. T_{air} (Air temperature) is taken as 20 °C. (Supply air 20 °C.)
A7. p_{atm} (Barometric pressure) is taken as 101.325 kPa. (Sea level.)
A8. v_{target} (Target velocity (velocity method)) is taken as 6 m/s. (Velocity method only; equal friction does not use it, so the template default is kept.)

4. Working

[10] ρ = p / (287.055 T)
= 101320 Pa / (287.055 × 293.15 K)
= 1.2041 kg/m³
Air density

[11] μ = Sutherland's law
= T = 293.15 K
= 18.133 uPa·s
Air dynamic viscosity

[12] a_{s1} = AHU to riser: side a
= as given
= 800 mm
Side a, AHU to riser

[13] b_{s1} = AHU to riser: side b
= from the target equivalent diameter 573.91 mm
= 400 mm
Side b, AHU to riser

[14] De_{s1} = AHU to riser: equivalent diameter [1](Ch. 21)
= 1.30 (ab)^0.625/(a + b)^0.25
= 609.35 mm
Equivalent diameter, AHU to riser

[15] v_{s1} = AHU to riser: Q / A
= 2 / 0.32
= 6.25 m/s
Velocity, AHU to riser

[16] Re_{s1} = AHU to riser: ρ v_e D_e / μ (v_e = 4Q/πD_e^2)
= 1.2041 × 6.8581 × 0.60935 / 0.000018133
= 277500
Reynolds number, AHU to riser

[17] f_{s1} = AHU to riser: friction factor
= Colebrook–White
= 0.015999
Friction factor, AHU to riser

[18] dpL_{s1} = AHU to riser: f ρ v_e^2 / (2 D_e)
= 0.015999 × 1.2041 × 6.8581² / (2 × 0.60935)
= 0.74347 Pa/m

Friction rate, AHU to riser

[19] $dp_{f,s1}$ = AHU to riser: $(\Delta p/L)$ L
= 0.74347 × 15
= 11.152 Pa

Friction loss, AHU to riser

[20] $dp_{fit,s1}$ = AHU to riser: $\Sigma C\ \rho\ v^2/2$
= 0.8 × 23.518
= 18.814 Pa

Fitting loss, AHU to riser

[21] dp_{s1} = AHU to riser: friction + fittings + equipment
= 11.152 + 18.814 + 0
= 29.966 Pa

Section loss, AHU to riser (index run)

[22] D_{s2} = riser: diameter
= target 573.91 mm rounded up to 50 mm
= 600 mm

Diameter, riser

[23] v_{s2} = riser: Q / A
= 2 / 0.28274
= 7.0736 m/s

Velocity, riser

[24] Re_{s2} = riser: $\rho\ v_e\ D_e$ / μ ($v_e = 4Q/\pi D_e^2$)
= 1.2041 × 7.0736 × 0.6 / 0.000018133
= 281820

Reynolds number, riser

[25] f_{s2} = riser: friction factor
= Colebrook–White
= 0.015985

Friction factor, riser

[26] dpL_{s2} = riser: $f\ \rho\ v_e^2$ / $(2\ D_e)$
= 0.015985 × 1.2041 × 7.0736² / (2 × 0.6)
= 0.80253 Pa/m

Friction rate, riser

[27] $dp_{f,s2}$ = riser: $(\Delta p/L)$ L
= 0.80253 × 12
= 9.6303 Pa

Friction loss, riser

[28] $dp_{fit,s2}$ = riser: $\Sigma C\ \rho\ v^2/2$
= 0.3 × 30.124
= 9.0371 Pa

Fitting loss, riser

[29] dp_{s2} = riser: friction + fittings + equipment
= 9.6303 + 9.0371 + 0
= 18.667 Pa

Section loss, riser (index run)

[30] a_{s3} = floor main: side a
= as given
= 500 mm

Side a, floor main

[31] b_{s3} = floor main: side b
= from the target equivalent diameter 473.16 mm
= 400 mm

Side b, floor main

[32] De_{s3} = floor main: equivalent diameter [1](Ch. 21)
= 1.30 (ab)^0.625/(a + b)^0.25
= 488.12 mm
Equivalent diameter, floor main
[33] v_{s3} = floor main: Q / A
= 1.2 / 0.2
= 6 m/s
Velocity, floor main
[34] Re_{s3} = floor main: ρ v_e D_e / μ (v_e = 4Q/πD_e^2)
= 1.2041 × 6.4127 × 0.48812 / 0.000018133
= 207850
Reynolds number, floor main
[35] f_{s3} = floor main: friction factor
= Colebrook–White
= 0.016905
Friction factor, floor main
[36] dpL_{s3} = floor main: f ρ v_e^2 / (2 D_e)
= 0.016905 × 1.2041 × 6.4127² / (2 × 0.48812)
= 0.85744 Pa/m
Friction rate, floor main
[37] $dp_{f,s3}$ = floor main: (Δp/L) L
= 0.85744 × 25
= 21.436 Pa
Friction loss, floor main
[38] $dp_{fit,s3}$ = floor main: ΣC ρ v^2/2
= 1.1 × 21.674
= 23.841 Pa
Fitting loss, floor main
[39] dp_{s3} = floor main: friction + fittings + equipment
= 21.436 + 23.841 + 0
= 45.277 Pa
Section loss, floor main (index run)
[40] D_{s4} = branch to VAV: diameter
= target 312.82 mm rounded up to 50 mm
= 350 mm
Diameter, branch to VAV
[41] v_{s4} = branch to VAV: Q / A
= 0.4 / 0.096211
= 4.1575 m/s
Velocity, branch to VAV
[42] Re_{s4} = branch to VAV: ρ v_e D_e / μ (v_e = 4Q/πD_e^2)
= 1.2041 × 4.1575 × 0.35 / 0.000018133
= 96625
Reynolds number, branch to VAV
[43] f_{s4} = branch to VAV: friction factor
= Colebrook–White
= 0.01938
Friction factor, branch to VAV
[44] dpL_{s4} = branch to VAV: f ρ v_e^2 / (2 D_e)
= 0.01938 × 1.2041 × 4.1575² / (2 × 0.35)
= 0.57623 Pa/m
Friction rate, branch to VAV

[45] $dp_{f,s4}$ = branch to VAV: (Δp/L) L
= 0.57623 × 10
= 5.7623 Pa
Friction loss, branch to VAV
[46] $dp_{fit,s4}$ = branch to VAV: ΣC ρ v²/2
= 0.9 × 10.406
= 9.3657 Pa
Fitting loss, branch to VAV
[47] dp_{s4} = branch to VAV: friction + fittings + equipment
= 5.7623 + 9.3657 + 60
= 75.128 Pa
Section loss, branch to VAV (index run)
[48] D_{s5} = other branch: diameter
= target 406.05 mm rounded up to 50 mm
= 450 mm
Diameter, other branch
[49] v_{s5} = other branch: Q / A
= 0.8 / 0.15904
= 5.0301 m/s
Velocity, other branch
[50] Re_{s5} = other branch: ρ v_e D_e / μ (v_e = 4Q/πD_e²)
= 1.2041 × 5.0301 × 0.45 / 0.000018133
= 150310
Reynolds number, other branch
[51] f_{s5} = other branch: friction factor
= Colebrook–White
= 0.017809
Friction factor, other branch
[52] dpL_{s5} = other branch: f ρ v_e² / (2 D_e)
= 0.017809 × 1.2041 × 5.0301² / (2 × 0.45)
= 0.60286 Pa/m
Friction rate, other branch
[53] $dp_{f,s5}$ = other branch: (Δp/L) L
= 0.60286 × 18
= 10.851 Pa
Friction loss, other branch
[54] $dp_{fit,s5}$ = other branch: ΣC ρ v²/2
= 1 × 15.233
= 15.233 Pa
Fitting loss, other branch
[55] dp_{s5} = other branch: friction + fittings + equipment
= 10.851 + 15.233 + 60
= 86.084 Pa
Section loss, other branch
[56] dp_{index} = Σ section losses on the index run
= 4 section(s)
= 169.04 Pa
Index-run pressure drop
[57] dp_{all} = Σ section losses, all sections
= 5 section(s)
= 255.12 Pa
Sum over every section (not a path)

[58] v_{peak} = highest section velocity
= v of riser, the highest of 5 section(s)
= 7.0736 m/s
Highest velocity
[59] $u_{dp} = dp_{index} / dp_{limit}$
= 169.04 / 250
= 0.67616
Pressure-drop utilisation
[60] $u_v = v_{peak} / v_{limit}$
= 7.0736 / 8
= 0.88419
Velocity utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
dp_{index}	169.0 Pa	Index-run Δp
v_{peak}	7.074 m/s	Highest velocity (governing)
a_{s1}	800 mm	a, AHU to riser
b_{s1}	400 mm	b, AHU to riser
De_{s1}	609.3 mm	D_e, AHU to riser
v_{s1}	6.250 m/s	v, AHU to riser
Re_{s1}	277500	Re, AHU to riser
f_{s1}	0.015999	f, AHU to riser
dpL_{s1}	0.7435 Pa/m	Δp/L, AHU to riser
$dp_{f,s1}$	11.15 Pa	Friction loss, AHU to riser
$dp_{fit,s1}$	18.81 Pa	Fitting loss, AHU to riser
dp_{s1}	29.97 Pa	Section loss, AHU to riser
D_{s2}	600 mm	D, riser
v_{s2}	7.074 m/s	v, riser
Re_{s2}	281820	Re, riser
f_{s2}	0.015985	f, riser
dpL_{s2}	0.8025 Pa/m	Δp/L, riser
$dp_{f,s2}$	9.630 Pa	Friction loss, riser
$dp_{fit,s2}$	9.037 Pa	Fitting loss, riser
dp_{s2}	18.67 Pa	Section loss, riser
a_{s3}	500 mm	a, floor main
b_{s3}	400 mm	b, floor main
De_{s3}	488.1 mm	D_e, floor main
v_{s3}	6 m/s	v, floor main
Re_{s3}	207850	Re, floor main

SYMBOL	VALUE	DESCRIPTION
f_{s3}	0.016905	f, floor main
dpL_{s3}	0.8574 Pa/m	Δp/L, floor main
$dp_{f,s3}$	21.44 Pa	Friction loss, floor main
$dp_{fit,s3}$	23.84 Pa	Fitting loss, floor main
dp_{s3}	45.28 Pa	Section loss, floor main
D_{s4}	350 mm	D, branch to VAV
v_{s4}	4.158 m/s	v, branch to VAV
Re_{s4}	96625	Re, branch to VAV
f_{s4}	0.01938	f, branch to VAV
dpL_{s4}	0.5762 Pa/m	Δp/L, branch to VAV
$dp_{f,s4}$	5.762 Pa	Friction loss, branch to VAV
$dp_{fit,s4}$	9.366 Pa	Fitting loss, branch to VAV
dp_{s4}	75.13 Pa	Section loss, branch to VAV
D_{s5}	450 mm	D, other branch
v_{s5}	5.030 m/s	v, other branch
Re_{s5}	150310	Re, other branch
f_{s5}	0.017809	f, other branch
dpL_{s5}	0.6029 Pa/m	Δp/L, other branch
$dp_{f,s5}$	10.85 Pa	Friction loss, other branch
$dp_{fit,s5}$	15.23 Pa	Fitting loss, other branch
dp_{s5}	86.08 Pa	Section loss, other branch
dp_{all}	255.1 Pa	Σ all sections

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Highest velocity against limit	7.074 m/s	8 m/s	0.88	
PASS	Index-run pressure drop against limit	169.0 Pa	250 Pa	0.68	

7. Applicability

None.

8. Verification

Template bs.duct-sizing@1.2.1 Content hash a0afb84d934b74eb Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

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3. British Standards Institution. Ventilation for Buildings—Sheet Metal Air Ducts and Fittings With Rectangular Cross-Section—Dimensions. BS EN 1505. BSI.
4. British Standards Institution. Ventilation for Buildings—Sheet Metal Air Ducts and Fittings With Circular Cross-Section—Dimensions. BS EN 1506. BSI.